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7.24

```
MariaDB [(none)]> create database company;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> use company;
Database changed

MariaDB [company]> CREATE TABLE Department (deptNo INT PRIMARY KEY,
deptName VARCHAR(50) NOT NULL, mgrEmpNo INT);
Query OK, 0 rows affected (0.035 sec)

MariaDB [company]> create table Employee(empNo VARCHAR(5) PRIMARY KEY,
fName varchar(30) not null, lName varchar(30) not null, address
varchar(100), DOB DATE, sex CHAR(1) check (sex in ('M','F')), position
varchar(30) check (position in ('Manager', 'Team Leader', 'Analyst',
'Software Developer')), deptNo varchar(5), FOREIGN KEY (deptNo)
REFERENCES Department(deptNo));
Query OK, 0 rows affected (0.029 sec)

MariaDB [company]> CREATE TABLE Project (projNo VARCHAR(10) PRIMARY
KEY, projName VARCHAR(50) NOT NULL,
    -> deptNo VARCHAR(5), FOREIGN KEY (deptNo) REFERENCES
Department(deptNo));
Query OK, 0 rows affected (0.029 sec)

MariaDB [company]> CREATE TABLE WorksOn (empNo VARCHAR(5), projNo
VARCHAR(10), dateWorked DATE, hoursWorked INT CHECK (hoursWorked
BETWEEN 0 AND 40), PRIMARY KEY(empNo, projNo, dateWorked), FOREIGN
KEY(empNo) REFERENCES Employee(empNo), FOREIGN KEY(projNo) REFERENCES
Project(projNo));
Query OK, 0 rows affected (0.030 sec)

MariaDB [company]> ALTER TABLE Department ADD CONSTRAINT FK_Manager
FOREIGN KEY (mgrEmpNo) REFERENCES Employee(empNo);
Query OK, 0 rows affected (0.054 sec)
Records: 0  Duplicates: 0  Warnings: 0

MariaDB [company]> INSERT INTO Department VALUES ('D1','Human
Resources',NULL), ('D2','Information Technology',NULL),
('D3','Finance',NULL), ('D4','Marketing',NULL);
Query OK, 4 rows affected (0.001 sec)
Records: 4  Duplicates: 0  Warnings: 0

MariaDB [company]> select * from Department;
```

```
+--------+------------------------+----------+
| deptNo | deptName               | mgrEmpNo |
+--------+------------------------+----------+
| D1     | Human Resources        | NULL     |
| D2     | Information Technology | NULL     |
| D3     | Finance                | NULL     |
| D4     | Marketing              | NULL     |
+--------+------------------------+----------+
4 rows in set (0.000 sec)

MariaDB [company]> INSERT INTO Employee
    -> VALUES
    -> ('E1','John','Mensah','Accra','1995-06-10','M','Manager','D2'),
    ->
    -> ('E2','Mary','Owusu','Tema','1998-08-15','F','Software
Developer','D2'),
    ->
    ->
('E3','Daniel','Boateng','Kumasi','1997-03-20','M','Manager','D3'),
    ->
    ->
('E4','Sarah','Asante','Takoradi','1999-01-25','F','Manager','D1'),
    ->
    -> ('E5','Michael','Aidoo','Cape Coast','2000-11-30','M','Software
Developer','D2'),
    ->
    -> ('E6','Linda','Kusi','Accra','1996-04-17','F','Manager','D4');
Query OK, 6 rows affected (0.002 sec)
Records: 6  Duplicates: 0  Warnings: 0

MariaDB [company]> select * from Employee;
+-------+---------+---------+------------+------------+------
+--------------------+--------+
| empNo | fName   | lName   | address    | DOB        | sex  |
position           | deptNo |
+-------+---------+---------+------------+------------+------
+--------------------+--------+
| E1    | John    | Mensah  | Accra      | 1995-06-10 | M    | Manager
| D2     |
| E2    | Mary    | Owusu   | Tema       | 1998-08-15 | F    |
Software Developer | D2     |
| E3    | Daniel  | Boateng | Kumasi     | 1997-03-20 | M    | Manager
| D3     |
| E4    | Sarah   | Asante  | Takoradi   | 1999-01-25 | F    | Manager
| D1     |
| E5    | Michael | Aidoo   | Cape Coast | 2000-11-30 | M    |
Software Developer | D2     |
| E6    | Linda   | Kusi    | Accra      | 1996-04-17 | F    | Manager
| D4     |
+-------+---------+---------+------------+------------+------
```

```
+--------------------+---------+
6 rows in set (0.000 sec)

MariaDB [company]> UPDATE Department SET mgrEmpNo='E4' WHERE
deptNo='D1';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [company]> UPDATE Department SET mgrEmpNo='E1' WHERE
deptNo='D2';
Query OK, 1 row affected (0.000 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [company]> UPDATE Department SET mgrEmpNo='E3' WHERE
deptNo='D3';
Query OK, 1 row affected (0.000 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [company]> UPDATE Department SET mgrEmpNo='E6' WHERE
deptNo='D4';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [company]> select * from Department;
+--------+------------------------+----------+
| deptNo | deptName               | mgrEmpNo |
+--------+------------------------+----------+
| D1     | Human Resources        | E4       |
| D2     | Information Technology | E1       |
| D3     | Finance                | E3       |
| D4     | Marketing              | E6       |
+--------+------------------------+----------+
4 rows in set (0.000 sec)

MariaDB [company]> INSERT INTO Project VALUES ('SCCS','Student Course
System','D2'), ('IMS','Inventory Management System','D3'),
('HRMS','Human Resource System','D1'), ('MKT','Marketing
Website','D4'), ('APP','Mobile Application','D2');
Query OK, 5 rows affected (0.003 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [company]> select * from Project;
+--------+-----------------------------+--------+
| projNo | projName                    | deptNo |
+--------+-----------------------------+--------+
| APP    | Mobile Application          | D2     |
| HRMS   | Human Resource System       | D1     |
| IMS    | Inventory Management System | D3     |
| MKT    | Marketing Website           | D4     |
| SCCS   | Student Course System       | D2     |
```

```
+---------+-----------------------------+---------+
5 rows in set (0.000 sec)

MariaDB [company]> INSERT INTO WorksOn VALUES
('E1','SCCS','2026-06-01',8), ('E2','SCCS','2026-06-01',7),
('E2','APP','2026-06-02',6), ('E3','IMS','2026-06-03',8),
('E4','HRMS','2026-06-02',7), ('E5','SCCS','2026-06-03',9),
('E5','APP','2026-06-04',8), ('E6','MKT','2026-06-05',6),
('E1','APP','2026-06-06',5), ('E6','MKT','2026-06-07',8);
Query OK, 10 rows affected (0.001 sec)
Records: 10  Duplicates: 0  Warnings: 0

MariaDB [company]> select * from WorksOn;
+-------+--------+------------+-------------+
| empNo | projNo | dateWorked | hoursWorked |
+-------+--------+------------+-------------+
| E1    | APP    | 2026-06-06 |           5 |
| E1    | SCCS   | 2026-06-01 |           8 |
| E2    | APP    | 2026-06-02 |           6 |
| E2    | SCCS   | 2026-06-01 |           7 |
| E3    | IMS    | 2026-06-03 |           8 |
| E4    | HRMS   | 2026-06-02 |           7 |
| E5    | APP    | 2026-06-04 |           8 |
| E5    | SCCS   | 2026-06-03 |           9 |
| E6    | MKT    | 2026-06-05 |           6 |
| E6    | MKT    | 2026-06-07 |           8 |
+-------+--------+------------+-------------+
10 rows in set (0.000 sec)
```

7.25

```
MariaDB [company]> CREATE VIEW FemaleManagerProjects AS SELECT
p.projNo, p.projName, d.deptName, e.empNo,
    -> e.fName, e.lName FROM Project p JOIN Department d ON
p.deptNo=d.deptNo JOIN Employee e
    -> ON d.mgrEmpNo=e.empNo WHERE e.sex='F' ORDER BY p.projNo;
Query OK, 0 rows affected (0.019 sec)

MariaDB [company]> select * from FemaleManagerProjects;
+--------+-----------------------+-----------------+-------+-------
+--------+
| projNo | projName              | deptName        | empNo | fName |
lName  |
+--------+-----------------------+-----------------+-------+-------
+--------+
| HRMS   | Human Resource System | Human Resources | E4    | Sarah |
Asante |
| MKT    | Marketing Website     | Marketing       | E6    | Linda |
Kusi   |
```

```
+--------+-----------------------+-----------------+-------+-------
+---------+
2 rows in set (0.002 sec)
```

7.26

```
MariaDB [company]> CREATE VIEW EmployeeProjectHours AS SELECT e.empNo,
e.fName, e.lName, p.projName, w.hoursWorked FROM Employee e JOIN
WorksOn w ON e.empNo=w.empNo JOIN Project p ON w.projNo=p.projNo;
Query OK, 0 rows affected (0.022 sec)

MariaDB [company]> select * from EmployeeProjectHours;
+-------+---------+---------+-----------------------------
+-------------+
| empNo | fName   | lName   | projName                    |
hoursWorked |
+-------+---------+---------+-----------------------------
+-------------+
| E1    | John    | Mensah  | Mobile Application          |
5 |
| E1    | John    | Mensah  | Student Course System       |
8 |
| E2    | Mary    | Owusu   | Mobile Application          |
6 |
| E2    | Mary    | Owusu   | Student Course System       |
7 |
| E3    | Daniel  | Boateng | Inventory Management System |
8 |
| E4    | Sarah   | Asante  | Human Resource System       |
7 |
| E5    | Michael | Aidoo   | Mobile Application          |
8 |
| E5    | Michael | Aidoo   | Student Course System       |
9 |
| E6    | Linda   | Kusi    | Marketing Website           |
6 |
| E6    | Linda   | Kusi    | Marketing Website           |
8 |
+-------+---------+---------+-----------------------------
+-------------+
10 rows in set (0.001 sec)
```

7.27

```
MariaDB [company]> create VIEW EmpProject(empNo, projNo, totalHours)
AS SELECT w.empNo, w.projNo, SUM(hoursWorked) FROM Employee e, Project
p, WorksOn w WHERE e.empNo = w.empNo AND p.projNo = w.projNo GROUP BY
w.empNo, w.projNo;
Query OK, 0 rows affected (0.024 sec)
```

```
MariaDB [company]> select * from EmpProject;
+-------+--------+------------+
| empNo | projNo | totalHours |
+-------+--------+------------+
| E1    | APP    |          5 |
| E1    | SCCS   |          8 |
| E2    | APP    |          6 |
| E2    | SCCS   |          7 |
| E3    | IMS    |          8 |
| E4    | HRMS   |          7 |
| E5    | APP    |          8 |
| E5    | SCCS   |          9 |
| E6    | MKT    |         14 |
+-------+--------+------------+
9 rows in set (0.001 sec)

MariaDB [company]> select projNo FROM EmpProject WHERE projNo =
'SCCS';
+--------+
| projNo |
+--------+
| SCCS   |
| SCCS   |
| SCCS   |
+--------+
3 rows in set (0.003 sec)

MariaDB [company]> select COUNT(projNo) from EmpProject WHERE empNo =
'E1';
+---------------+
| COUNT(projNo) |
+---------------+
|             2 |
+---------------+
1 row in set (0.001 sec)

MariaDB [company]> select empNo, totalHours FROM EmpProject GROUP BY
empNo;
+-------+------------+
| empNo | totalHours |
+-------+------------+
| E1    |          5 |
| E2    |          6 |
| E3    |          8 |
| E4    |          7 |
| E5    |          8 |
| E6    |         14 |
+-------+------------+
6 rows in set (0.001 sec)
```

7.28

I would maintain a materialized view by updating it whenever the Part table changes. For instance, on INSERT, I would add a partNo if partCost > 1000. On update, add or remove partNo if the cost crosses the 1000 threshold. On delete, remove a partNo only if no other record for that part has a part cost greater than 1000.